

# ISC 2026 EXAMINATION

## Sample Question Paper - 4

### Physics

**Time Allowed: 3 hours and 15 minutes**

**Maximum Marks: 70**

**General Instructions:**

- You are allowed an additional 15 minutes for only reading the question paper.
- You must NOT start writing during reading time.
- This question paper has 20 questions.
- The paper has four sections: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
- Section A consists of one question having fourteen sub-parts of one mark each.
- Section B consists of seven questions of two marks each.
- Section C consists of nine questions of three marks each.
- Section D consists of three questions of five marks each.
- Answer all questions.
- The intended marks for questions are given in brackets [].
- A list of useful constants and relations is given at the end of this paper.
- A simple scientific calculator without a programmable memory may be used for calculations.

**Section A**

1. **Answer the following questions:** [14]

(a) **In questions (i) to (vii) below, choose the correct alternative (a), (b), (c) or (d) for each of the questions given below:**

i. Specific resistance of the material of a wire depends on: [1]

- |                             |                                       |
|-----------------------------|---------------------------------------|
| a) Volume of the wire.      | b) Area of cross section of the wire. |
| c) Temperature of the wire. | d) Length of the wire.                |

ii. A wire of length 2 m is bent to form a circular loop of single turn. If 1 A current flows through the loop the magnetic moment is [1]

- |                    |                    |
|--------------------|--------------------|
| a) $\frac{2}{\pi}$ | b) $\frac{1}{\pi}$ |
| c) $\frac{3}{\pi}$ | d) $\pi$           |

iii. Proper arrangement of Gamma rays, Microwave, IR wave and UV rays in ascending order of frequency is [1]

- |                                               |                                               |
|-----------------------------------------------|-----------------------------------------------|
| a) IR rays > UV rays > Microwave > Gamma rays | b) Microwave > IR rays > UV rays > Gamma rays |
| c) UV rays > Gamma rays > Microwave > IR rays | d) Gamma rays > UV rays > IR rays > Microwave |

A Rowland ring of mean radius 15 cm has 3500 turns of wire wound on a ferromagnetic core of

relative permeability 800. What is the magnetic field  $B$  in the core for a magnetising current of 1.2 A?

3. A radio can tune into any station in the 7.5 MHz to 12 MHz band. What is its corresponding wavelength? [2]
4. An AC generator consists of a coil of 1000 turns and cross-sectional area of  $100 \text{ cm}^2$ , rotating at an angular speed of 100 rpm in a uniform magnetic field of  $3.6 \times 10^{-2} \text{ T}$ . Calculate the maximum emf produced in the coil. [2]

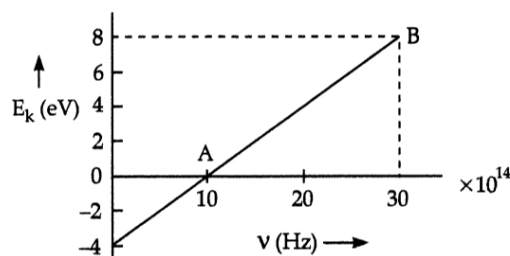
OR

An alternating voltage given by  $E = 140 \sin 314t$  is connected across a pure resistor of  $50 \Omega$ . Find

- i. the frequency of the source.
- ii. the rms current through the resistor.
5. Explain with the help of a circuit diagram the working of a photodiode. Write briefly, how it is used to detect the optical signals? [2]
6. A galvanometer coil has a resistance of  $15 \Omega$  and the meter shows full scale deflection for a current of 4 mA. How will you convert the meter into an ammeter of range 0 to 6 A? [2]
7. An ebonite rod held in hand can be charged by rubbing with flannel but a copper rod cannot be charged like this, why? [2]
8. Define the following terms used in electronic devices. [2]
  - i. Reverse breakdown voltage.
  - ii. V-I characteristic of forward biased diode.

### Section C

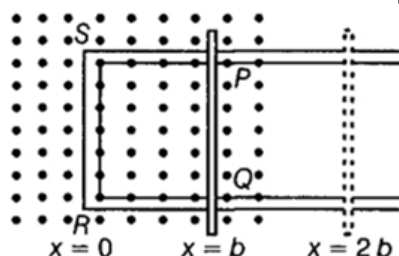
9. In an experiment of photoelectric effect, the graph of maximum kinetic energy  $E_k$  of the emitted photoelectrons versus the frequency  $\nu$  of the incident light is a straight line AB as shown in figure below: [3]



Find:

- i. Threshold frequency of the metal.
- ii. Work function of the metal.
- iii. Stopping potential for the photoelectrons emitted by the light of frequency  $\nu = 30 \times 10^{14} \text{ Hz}$ .
10. State Faraday's laws of electromagnetic induction. Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move in a uniform magnetic field  $\vec{B}$  perpendicular to the plane of the paper. The field extends from  $x = 0$  to  $x = b$  and is zero for  $x > b$ . Assume that only the arm PQ possesses resistance  $r$ . When the arm PQ is pulled outward from  $x = 0$  to  $x = 2b$  and is then moved backward to  $x = 0$  with constant speed  $v$ , obtain the expressions for the flux and the induced emf. [3]

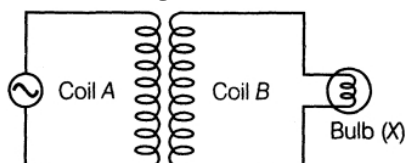
Sketch the variation of these quantities with distance  $0 \leq x \leq 2b$ .



OR

The figure given below shows an arrangement by which current flows through the bulb X connected with coil B, when AC is passed through coil A.

- Name the phenomenon involved.
- If a copper sheet is inserted in the gap between the coils, explain how the brightness of the bulb would change?



- Three equal charges of  $5.0\mu\text{C}$  each are placed at the three vertices of an equilateral triangle of side 5.0 cm each. Calculate the electrostatic potential energy of the system of charges. [3]
- Calculate mass defect and binding energy per nucleon of  ${}^{20}_{10}\text{Ne}$ , given [3]  
 Mass of  ${}^{20}_{10}\text{Ne} = 19.992397 \text{ u}$   
 Mass of  ${}^1_1\text{H} = 1.007825 \text{ u}$   
 Mass of  ${}^1_0\text{n} = 1.008665 \text{ u}$

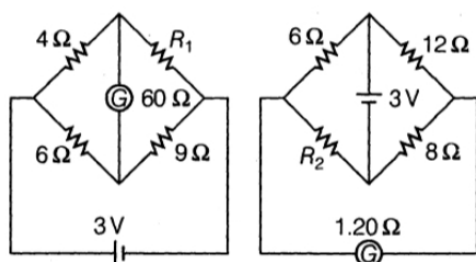
OR

Obtain the binding energy of the nuclei  ${}^{56}_{26}\text{Fe}$  and  ${}^{209}_{83}\text{Bi}$  in units of MeV from the following data:

$$m({}^{56}_{26}\text{Fe}) = 55.934939 \text{ u}$$

$$m({}^{209}_{83}\text{Bi}) = 208.980388 \text{ u}$$

- Radius of curvature of an equi-convex lens of glass ( $n = 1.5$ ) is 30 cm. Find its focal length. An object of height 5.0 cm is placed at a distance of 60 cm from the optical centre of the lens. Find the position and the height of the image formed. [3]
- Why is the binding energy per nucleon found to be constant for nuclei in the range of mass number ( $A$ ) lying between 30 and 170? [3]
  - When a heavy nucleus with mass number  $A = 240$  breaks into two nuclei,  $A = 120$ , energy is released in the process.
- Figure shows two circuits each having a galvanometer and a battery of 3 V. When the galvanometer in each arrangement do not show any deflection, obtain the ratio  $\frac{R_1}{R_2}$ . [3]



- Explain the following: [3]
  - Diamagnetism is the result of induced magnetic dipole moments.

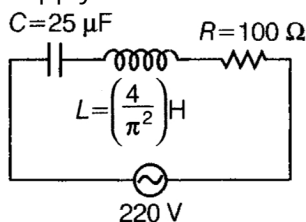
ii. Hysteresis associated with a loss in electromagnetic energy.

17. Suppose that the electric field amplitude of an electromagnetic wave is  $E_0 = 120 \text{ N/C}$  and that its frequency is  $\nu = 50.0 \text{ MHz}$ . [3]

- Determine  $B_0$ ,  $\omega$ ,  $k$  and  $\lambda$ .
- Find expressions for  $\mathbf{E}$  and  $\mathbf{B}$ .

#### Section D

18. i. Figure below shows a capacitor  $C$ , an inductor  $L$  and a resistor  $R$  connected in series to an AC supply of  $220 \text{ V}$ . [5]



Calculate

- the resonant frequency of the given L-C-R circuit.
  - current flowing through the circuit.
  - average power consumed by the circuit.
- ii. In a series L-C-R circuit, what is the phase difference between  $V_L$  and  $V_C$ , where  $V_L$  is the potential difference across the inductor and  $V_C$  is the potential difference across the capacitor?

OR

- i. A coil having self-inductance of  $0.7 \text{ H}$  and resistance of  $165 \Omega$  is connected to an a.c. source of  $275 \text{ V}$ ,  $50 \text{ Hz}$ . if  $\pi = \frac{22}{7}$ , then Calculate:

- Reactance of the coil
- Impedance of the coil
- Current flowing through the coil.

- ii. Draw a labelled graph showing variation of impedance of a series L-C-R circuit with frequency of the AC supply.

19. In Young's double slit experiment, the two slits  $0.15 \text{ mm}$  apart are illuminated by monochromatic light of wavelength  $450 \text{ nm}$ . The screen is  $1 \text{ m}$  away from the slits. Find the distance of the second [5]

- bright fringe.
- dark fringe from the central maxima

OR

Light waves from coherent sources arrive at two points on a screen with path difference of  $0$  and  $\lambda/2$ . Find the ratio of intensities at the points.

20. i. Using Ampere's circuital law, derive the expression for the magnetic field in the vector form at a point on the axis of a solenoid. [5]
- ii. What does a toroid consist of? Find out the expression for the magnetic field inside a toroid for  $N$  turns of the coil having the average radius  $r$  and carrying a current  $I$ . Show that the magnetic field in the open space interior and exterior to the toroid is zero.

# Solution

## Section A

1. Answer the following questions:

(a) In questions (i) to (vii) below, choose the correct alternative (a), (b), (c) or (d) for each of the questions given below:

- i. **(c)** Temperature of the wire.

**Explanation:**

Specific resistance depends on the temperature and the nature of the material.

- ii. **(b)**  $\frac{1}{\pi}$

**Explanation:**

Perimeter of the loop =  $2 \pi r = 2$

$$\therefore r = \frac{1}{\pi}$$

$$\therefore \text{Area} = \pi r^2 = \therefore \times \left(\frac{1}{\pi}\right)^2 = \frac{1}{\pi}$$

$$\text{Magnetic dipole moment} = NiA = 1 \times 1 \times \left\{\frac{1}{\pi}\right\} = \frac{1}{\pi}$$

- iii. **(d)** Gamma rays > UV rays > IR rays > Microwave

**Explanation:**

Frequency range of Gamma rays:  $10^{22}$  -  $10^{19}$  Hz

Frequency range of UV rays:  $10^{17}$  -  $10^{15}$  Hz

Frequency range of IR rays:  $10^{14}$  -  $10^{12}$  Hz

Frequency range of Microwave:  $10^{13}$  -  $10^9$  Hz

- iv. **(a)**  $-X_g$

**Explanation:**

As internal resistance of generator is already equal to the external resistance  $R_g$ . So to deliver maximum power, i.e., to make reactance equal to zero, the reactance in external circuit will be  $-X_g$ . In order to deliver maximum power, the generator to the load, the total reactance must be equal to zero, i.e.,  $X_L + X_g = 0$ ,  $X_L = -X_g$ .

- v. **(c)**  $(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$

**Explanation:**

From above mentioned three given elements, the energy band gap of carbon is the maximum and that of germanium is the least. The energy band gaps of these elements are related as:

$$(E_g)_C > (E_g)_{Si} > (E_g)_{Ge}$$

- vi. **(d)**  $\frac{\mu NI}{l}$

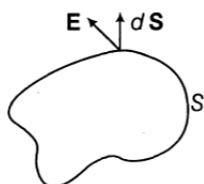
**Explanation:**

$$B = \mu_0 n I = \frac{\mu_0 N I}{l} \text{ where } n = \frac{N}{l}$$

- vii. **(a)** the total flux through the surface is zero

**Explanation:**

This means total charge and so total flux is zero.



$\oint E \cdot dS$  = Total flux through the surface.

Given, total flux through surface =  $\oint E \cdot dS = 0 \dots(i)$

By Gauss's law (total flux through) =  $\frac{1}{\epsilon_0}$  (net charge enclosed in S) ...(ii)

From Eqs. (i) and (ii), we get

Net charge inside S = 0

Total flux through the surface S is zero.

(b) Answer the following questions briefly:

i. We know,  $\beta = \frac{D\lambda}{d}$

As wavelength increases, fringe width increases. Since,  $\lambda_{\text{red}} > \lambda_{\text{violet}}$ , the fringe width increases.

ii. In our daily life, we deal with bigger (macroscopic) particles and their wavelength is of the order of  $10^{-35}$  m or less. Hence, wave effect is insignificant.

iii. Soft iron is preferred because in soft iron, energy loss per unit volume due to hysteresis is low.

iv. Electric field is always normal to the equipotential surface at every point, because no work is done, as

$$W = q_0(V_A - V_B)$$

$$\Rightarrow V_A - V_B = 0$$

hence  $W = 0$ .

If the field were not normal to the equipotential surface, it would have a non-zero component along the surface. So, to move a test charge against this component, a work would have to be done.

v.  $H_\alpha$ -line of the Balmer series in the emission spectrum of H-atom is obtained in visible region, because

$$\frac{1}{\lambda} = R \left( \frac{1}{2^2} - \frac{1}{n^2} \right)$$

where,  $n = 3, 4, 5, \dots$

vi. Wavelength of violet light is smaller than that of red light. Also, angle of minimum deviation,  $\delta_m = (\mu - 1) A$

$$\Rightarrow \delta_m \propto \mu$$

$$\text{As, } \mu_R < \mu_V$$

$$\Rightarrow (\delta_m)_R < (\delta_m)_V$$

As deviation is less for red light, hence angle of deviation decreases.

vii. Nuclear fusion involves the fusion of two or more very light nuclei to form a single nucleus.

Since the nuclei to be fused are positively charged, they would repel each other strongly.

Thus, they must be brought close together by high pressure as well as high kinetic energies.

For this, very high temperatures of the order of  $10^9$  K are required. since very high

temperatures are required for nuclear fusion, it is also called thermo-nuclear reaction.

### Section B

2. The horizontal and vertical components of the earth's magnetic field are perpendicular to each other.

Horizontal component of earth's magnetic field,

$$H = B_e \cos 60^\circ = B \text{ (given)}$$

$$\Rightarrow B_e \times \frac{1}{2} = B \text{ or } B_e = 2B$$

Vertical component of earth's magnetic field,

$$V = B_e \sin 60^\circ \Rightarrow V = 2B \times \frac{\sqrt{3}}{2} \Rightarrow B_e = \sqrt{3}B$$

OR

Given, radius of Rowland ring,  $r = 15 \text{ cm} = 0.15 \text{ m}$

Number of turns,  $N = 350$

Relative permeability of ferromagnetic core,  $\mu_r = 800$

Current,  $I = 1.2 \text{ A}$

Magnetic field due to the toroid,

$$B = \mu n I \left[ \because n = \frac{\text{number of turns (N)}}{\text{length}} \right]$$

$$= \mu_0 \mu_r \frac{N}{2\pi r} \cdot I \quad [\because \text{length of toroid} = 2\pi r]$$

$$= 4 \times 3.14 \times 10^{-7} \times 800 \times \frac{3500 \times 1.2}{2 \times 3.14 \times 0.15} = 4.48 \text{ T}$$

3. For 7.5 MHz band,

$$\text{Wavelength, } \lambda_1 = \frac{c}{v} = \frac{3 \times 10^8}{7.5 \times 10^6} = 40 \text{ m}$$

For 12 MHz band,

$$\text{Wavelength, } \lambda_2 = \frac{c}{v} = \frac{3 \times 10^8}{12 \times 10^6} = 25 \text{ m}$$

So, wavelength range is from 25 m to 40 m.

4. Given,  $N = 1000$ ,

$$A = 100 \text{ cm}^2 = 10^{-2} \text{ m}^2,$$

$$v = 100 \text{ rpm} = \frac{100}{60} \text{ rps},$$

$$B = 3.6 \times 10^{-2} \text{ T, } e_0 = ?$$

$\therefore$  Maximum emf produced in the coil is

$$e_0 = NBA \omega$$

$$= NBA (2\pi v)$$

$$= 1000 \times 3.6 \times 10^{-2} \times 10^{-2} \times 2 \times \frac{22}{7} \times \frac{100}{60}$$

$$= 3.77 \text{ V}$$

OR

i. As given,  $E = 140 \sin 314t$

On comparing with  $E = E_0 \sin \omega t$ , we have

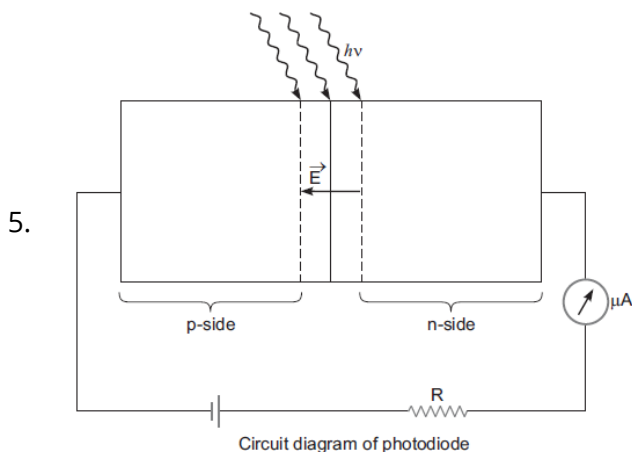
$$\omega = 314, E_0 = 140 \text{ V}$$

$$\therefore \omega = 2\pi v \Rightarrow v = \frac{\omega}{2\pi} = \frac{314}{2 \times 3.14} = 50 \text{ Hz}$$

ii.  $E_0 = 140 \text{ V}$

$$E_{\text{rms}} = \frac{E_0}{\sqrt{2}} = \frac{140}{\sqrt{2}} = 99.29 \text{ V}$$

$$\therefore I_{\text{rms}} = \frac{E_{\text{rms}}}{R} = \frac{99.29}{50} = 1.98 \text{ A}$$



Working: In diode (any type of diode), an electric field 'E' exists across the junction from n-side to p-side, when light with energy  $h\nu$  greater than energy gap  $E_g$  illuminates the junction. Then electron-hole pairs are generated due to absorption of photons, in or near the depletion region of the diode. Due to existing electric field, electrons and holes get separated. The free electrons are collected on n-side and holes are collected on p-side, giving rise to an emf.

Due to this generated emf, an electric current of  $\mu\text{A}$  order flows through the external

resistance. Detection of Optical Signals: It is easier to observe the change in the current with change in the light intensity if a reverse bias is applied. Thus, photodiode can be used as a photodetector to detect optical signals.

6.  $G = 15\Omega$

Current for which, the galvanometer shows full scale deflection,  $i = 4\text{mA}$ .

Range of ammeter is  $4\text{ mA}$ , which needs to be converted to  $6\text{A}$ .

$\therefore I = 6\text{A}$

A short resistance  $R$  is to be connected in parallel to the galvanometer.

$$R = \frac{iG}{(I-i)} = 10\text{m}\Omega$$

7. Both the human body and the copper rod conduct electricity. When it is attempted to charge a copper rod by rubbing, the charge flows from the rod to the earth through the hand. However, when ebonite rod is charged by rubbing, the charges so produced stay on the ebonite rod as it is a bad conductor of electricity.

8. i. Reverse breakdown, also known as reverse breakdown voltage, is the voltage at which a diode or other semiconductor device begins to conduct in the reverse direction.  
 ii. The forward V-I characteristic of the P-N junction diode shows that the current grows exceptionally slowly at the beginning. The curve is nonlinear because the external voltage delivered to the p-n junction is used to overcome the potential barrier in this region.

### Section C

9. i. Threshold frequency  $(\nu_0) = 10 \times 10^{14}\text{ Hz}$

ii.  $W = (h\nu_0)$

$$= 6.6 \times 10^{-34} \times 10 \times 10^{14}$$

$$= 6.6 \times 10^{-19}\text{ J} = 4.125\text{ eV}$$

iii.  $E_K = 8 \times 1.6 \times 10^{-19}\text{ J} = 8\text{ eV}$

$$V_s = \frac{8 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-19}}$$

$$V_s = 8.0\text{ V}$$

10. Faraday's First Law : The changing magnetic field linked with a conductor induces an electromotive force in the conductor.

Faraday's Second Law : The induced electromotive force is proportional to the rate of change of magnetic field linked with the conductor.

Faraday's law of electromagnetic induction states that whenever there is a change in magnetic flux linked with of a coil, an emf is induced in the coil. The induced emf is proportional to the rate of change of magnetic flux linked with the coil. i.e.,  $e \propto \frac{\Delta\phi}{\Delta t}$

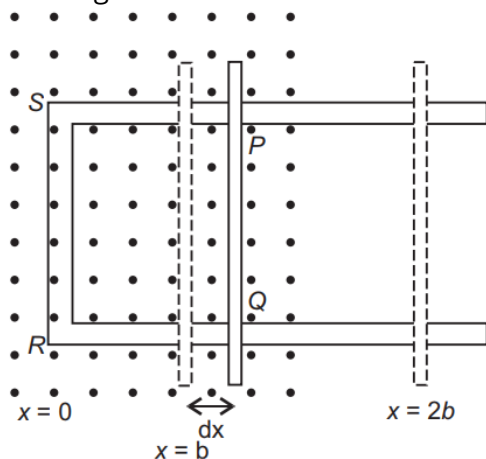
If the coil contains  $N$ -turns, then  $e = -N \left( \frac{\Delta\phi}{\Delta t} \right)$

Let length of conductor  $PQ = l$  As  $x = 0$ , magnetic flux  $\phi = 0$ .

When  $PQ$  moves a small distance from  $x$  to  $x + dx$ , then magnetic flux linked  $= B dA = B l dx$

The magnetic field is from  $x = 0$  to  $x = b$ , so final magnetic flux  $= \sum B l dx = B l \sum dx = B l b$  (increasing) Mean magnetic flux from  $x = 0$  to  $x = b$  is  $\frac{1}{2} B l b$ .

The magnetic flux from  $x = b$  to  $x = 2b$  is zero

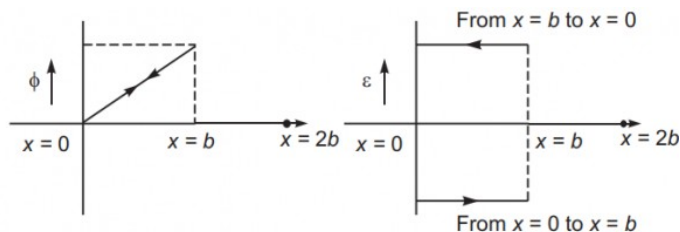


$$= -Bl \frac{dx}{dt} = -Blv$$

where  $v = \frac{dx}{dt}$  = velocity of arm PQ from  $x = 0$  to  $x = b$ .

During return from  $x = 2b$  to  $x = b$ , the induced emf is zero; but now area is decreasing so magnetic flux is decreasing, and induced emf will be in opposite direction.  $\epsilon = Blv$  Graph is shown in figure.

$$\epsilon = Blv$$



OR

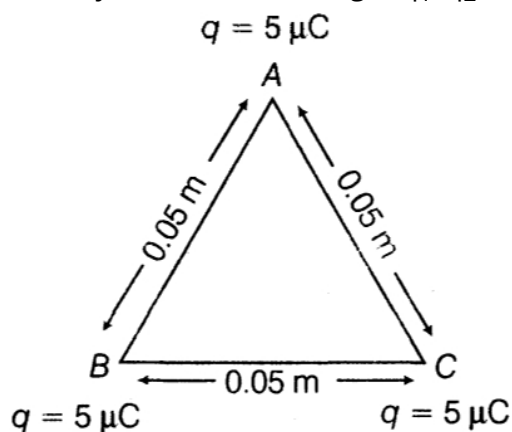
i. Mutual induction

ii. Due to magnetic induction bulb will glow and due to change in magnetic flux a induced current is produce in coil B so bulb will glow

11. Potential energy (PE) of a system of two charges  $q_1$  and  $q_2$  separated by  $r$  is

$$U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r}$$

For a system of three charges  $q_1$ ,  $q_2$  and  $q_3$  the total potential energy  $U$  is



$$U = U_{AB} + U_{BC} + U_{CA}$$

$$= \frac{1}{4\pi\epsilon_0} \left( \frac{q_1 q_2}{r_{12}} + \frac{q_2 q_3}{r_{23}} + \frac{q_1 q_3}{r_{13}} \right) = \frac{1}{4\pi\epsilon_0} \cdot 3 \frac{q^2}{r}$$

$$[\because r_{12} = r_{23} = r \text{ and } q_1 = q_2 = q_3 = q]$$

$$= 9 \times 10^9 \times \frac{5 \times 10^{-6} \times 5 \times 10^{-6}}{5 \times 10^{-2}} \times 3 = 13.5 \text{ J}$$

12. In a nucleus of  ${}^{20}_{10}\text{Ne}$

Number of protons = 10

Number of neutrons =  $20 - 10 = 10$

Total mass of 10 protons and 10 neutrons

$$= 10 m_p + 10 m_n = 10 (m_p + m_n)$$

$$= 10 \times (1.007825 + 1.008665)$$

$$= 10 \times 2.01649 \text{ u} = 20.1649 \text{ u}$$

$$\text{Mass defect, } \Delta m = 20.1649 - 19.992397$$

$$= 0.172503 \text{ u}$$

$$\text{Total binding energy} = 0.172503 \times 931$$

$$= 160.600293 \text{ MeV}$$

$$\text{Eb per nucleon} = \frac{160.600293}{20}$$

$$= 8.03001465 \text{ MeV/ nucleon}$$

OR

Given,  $m_p = 1.00783 \text{ u}$ ,  $m_n = 1.00867 \text{ u}$

i. For  ${}^{56}_{26}\text{Fe}$ , there are 26 protons and  $(56 - 26) = 30$  neutrons.

$$\Delta m = \text{mass of nucleons} - \text{mass of nucleus}$$

$$= 26m_p + 30m_n - m$$

$$= 26 \times 1.00783 + 30 \times 1.00867 - 55.934939 = 0.528741 \text{ u}$$

$$\text{Total binding energy} = \Delta m \times 931 \text{ MeV}$$

$$= 0.528741 \times 931$$

$$= 492.26 \text{ MeV}$$

Binding energy per nucleon

$$= \frac{\text{Binding energy}}{\text{Number of nucleons}}$$

$$= \frac{492.26}{56} = 8.790 \text{ MeV}$$

ii. For  ${}_{83}^{209}\text{Bi}$ , there are 83 protons and  $(209 - 83) = 126$  neutrons.

$$\Delta m = \text{mass of nucleons} - \text{mass of nucleus}$$

$$= 83 m_p + 126 m_n - m$$

$$\Delta m = 83 \times 1.00783 + 126 \times 1.00867 - 208.980388 = 1.761922 \text{ u}$$

$$\text{Binding energy} = \Delta m \times 931 \text{ MeV}$$

$$= 1.761922 \times 931 = 1640.35 \text{ MeV}$$

Binding energy per nucleon

$$= \frac{\text{Binding energy}}{\text{Number of nucleons}}$$

$$= \frac{1640.35}{209} = 7.848 \text{ MeV}$$

Thus, binding energy per nucleon of Fe is more than Bi.

13. Using Lens Maker's Formula

$$\frac{1}{f} = (n - 1) \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = (1.5 - 1) \left( \frac{1}{30} + \frac{1}{30} \right)$$

$$= \frac{0.5 \times 2}{30} = \frac{1}{30}$$

$$f = 30 \text{ cm}$$

$$m = \frac{f}{f+u} = \frac{30}{30-60} = \frac{30}{-30} = -1$$

$$\text{and } m = \frac{v}{u}$$

$$-1 = \frac{v}{-60}$$

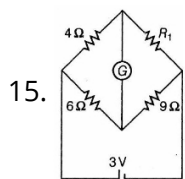
$$\therefore v = 60 \text{ cm}$$

$$|m| = \frac{h_i}{h_o}$$

$$1 = \frac{h_i}{5}$$

$$h_i = 5 \text{ cm}$$

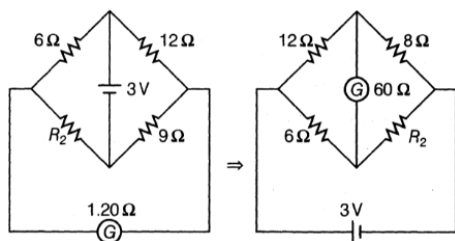
14. i. The binding energy per nucleon for nucleus of range,  $30 < A < 170$  is close to its maximum value. So, the nucleus belongs to this region is highly stable and does not show radioactivity.
- ii. Binding energy per nucleon is smaller for heavier nuclei than the middle ones, i.e. heavier nuclei are less stable. When a heavier nucleus such as a nucleus of mass number 240 splits into lighter nuclei (mass number 120), the BE/nucleon changes from about 7.6 MeV to 8.4 MeV. Greater BE of the product nuclei result in the liberation of energy.



For balanced Wheatstone bridge, there will be no deflection in the galvanometer.

$$\frac{4}{R_1} = \frac{6}{9}$$

$$\Rightarrow R_1 = \frac{4 \times 9}{6} = 6\Omega$$



For the equivalent circuit, when the Wheatstone bridge is balanced, there will be no deflection in the galvanometer.

$$\therefore \frac{12}{8} = \frac{6}{R_2}$$

$$\Rightarrow R_2 = \frac{6 \times 8}{12} = 4\Omega$$

$$\therefore \frac{R_1}{R_2} = \frac{6}{4} = \frac{3}{2}$$

16. i. Materials which lack permanent magnetic dipole moment are called diamagnetic. It is induced by a change in the orbital motion of electrons with applied magnetic field. Diamagnetic material exposed to an external magnetic field, magnetic moment is induced in a direction opposite to that of an applied magnetic field.
- ii. The hysteresis loss instead is the energy entrapped in magnetic materials exposed to a magnetic field in the form of residual magnetization - a typical behavior of FM materials - which is then lost as heat during the demagnetization step.

17. Given, amplitude of an electromagnetic wave,

$$E_0 = 120 \text{ N/C}$$

$$\text{Frequency of wave, } \nu = 50 \text{ MHz} = 50 \times 10^6 \text{ Hz}$$

- i. Speed of light in vacuum,

$$c = \frac{E_0}{B_0}$$

$$B_0 = \frac{E_0}{c} = \frac{120}{3 \times 10^8} = 40 \times 10^{-8}$$

$$= 400 \times 10^{-9} \text{ T}$$

$$= 400 \text{ nT}$$

Angular frequency of electromagnetic wave,

$$\omega = 2\pi\nu = 2 \times 3.14 \times 50 \times 10^6$$

$$\omega = 3.14 \times 10^8 \text{ rad/s}$$

Wave number of electromagnetic wave,

$$k = \frac{\omega}{c} = \frac{3.14 \times 10^8}{3 \times 10^8} = 1.05 \text{ rad/m}$$

Wavelength of electromagnetic wave,

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{50 \times 10^6} = 6.00 \text{ m}$$

- ii. Expression of electric field,  $E = E_0 \sin(kx - \omega t)$

$$E = 120 \sin(1.05x - 3.14 \times 10^8 t)$$

Expression of magnetic field B,

$$B = B_0 \sin(kx - \omega t)$$

$$B = 4 \times 10^{-7} \sin(1.05x - 3.14 \times 10^8 t)$$

### Section D

18. i. Given,  $E_{\text{rms}} = 220 \text{ V}$ ,  $C = 25 \mu\text{F} = 25 \times 10^{-6} \text{ F}$ ,

$$L = \frac{4}{\pi^2} \text{ H}, R = 100\Omega$$

- a. Resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2 \times 3.14 \times \sqrt{\frac{4}{\pi^2} \times 25 \times 10^{-6}}}$$

$$= \frac{1}{2 \times 3.14 \times \frac{2}{314} \times 5 \times 10^{-3}} = 50 \text{ Hz}$$

b. Impedance of the circuit is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \dots (i)$$

$$\text{We know that, } X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$= \frac{1}{2 \times 3.14 \times 50 \times 25 \times 10^{-6}} = 127.3 \Omega$$

$$X_L = \omega L = 2\pi f L = 2 \times 3.14 \times \frac{4}{\pi^2} \times 50 = 127.3 \Omega$$

Substituting the all values in Eq. (i), we get

$$Z = \sqrt{(100)^2 + (127.3 - 127.3)^2}$$

$$= \sqrt{(100)^2} = 100 \Omega$$

$$\text{We know that, } I_{\text{rms}} = \frac{E_{\text{rms}}}{Z} = \frac{220}{100} = 2.2 \text{ A}$$

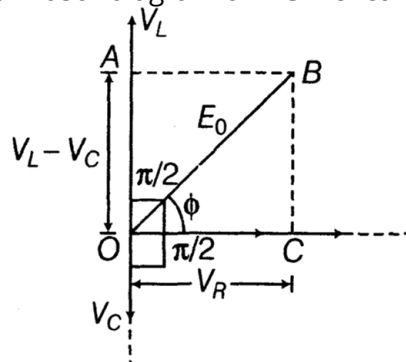
c. Average power consumed by the circuit

$$= E_{\text{rms}} \times I_{\text{rms}} \times \frac{R}{\sqrt{R^2 + \left(L\omega - \frac{1}{C\omega}\right)^2}}$$

$$\therefore Z = R = 100 \Omega$$

$$\text{So, power} = E_{\text{rms}} \times I_{\text{rms}} = 220 \times 2.2 = 484 \text{ W}$$

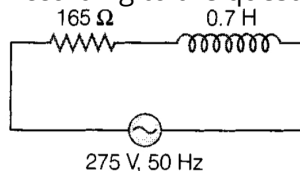
ii. Phasor diagram of L-C-R circuit is shown below.



The phase difference across the inductor and phase difference across the capacitor is  $180^\circ$ .

OR

i. According to the question,



a. Reactance of the coil is  $\omega L = X_L = (2\pi f L)$

$$= 2 \times \frac{22}{7} \times 50 \times 0.7$$

$$= 220 \Omega$$

b. Impedance of the coil =  $\sqrt{X_L^2 + R^2}$

$$= \sqrt{(\omega L)^2 + R^2}$$

$$= \sqrt{(2\pi \times 50 \times 0.7)^2 + (165)^2}$$

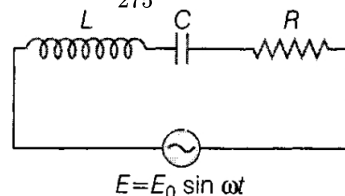
$$= \sqrt{(48400 + 27225)}$$

$$= \sqrt{75625} = 275$$

c. Current flowing through the coil,  $V = IR$

$$\Rightarrow I = \frac{V}{R} \text{ (where, } R = \text{resistance of circuit and } V = \text{voltage applied)}$$

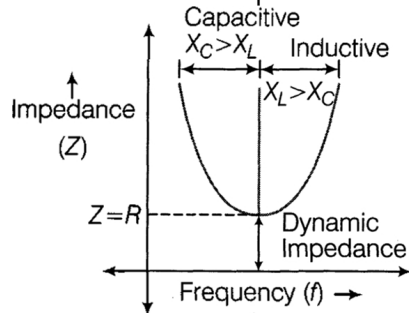
$$\Rightarrow I = \frac{275}{275} \Rightarrow I = 1 \text{ A}$$



$$\text{Impedance}(Z) = \sqrt{(X_L - X_C)^2 + R^2}$$

$$= \sqrt{\left(\omega L - \frac{1}{\omega C}\right)^2 + R^2}$$

ii. The variation of impedance of a series L-C-R circuit with frequency of AC supply is given below.



19. i. The distance of  $n$ th order bright fringe from central fringe is given by

$$y_n = \frac{nD\lambda}{d}$$

For second bright fringe,

$$y_2 = \frac{2D\lambda}{d}$$

$$= \frac{2 \times 1 \times 4.5 \times 10^{-7}}{1.5 \times 10^{-4}},$$

$$y_2 = 6 \times 10^{-3} \text{ m}$$

The distance of the second bright fringe

$$y_2 = 6 \text{ mm}$$

ii. The distance of  $n$ th order dark fringe from central fringe is given by.

$$y'_n = (2n - 1) \frac{D\lambda}{2d}$$

For second dark fringe,  $n = 2$

$$y'_n = (2 \times 2 - 1) \frac{D\lambda}{2d} = \frac{3D\lambda}{2d}$$

$$\Rightarrow y'_n = \frac{3}{2} \times \frac{1 \times 4.5 \times 10^{-7}}{1.5 \times 10^{-4}}$$

$$= 4.5 \times 10^{-3} \text{ m}$$

$\therefore$  The distance of the second dark fringe

$$y'_n = 4.5 \text{ mm}$$

OR

$$\text{As, } I = I_0 \cos^2 \phi$$

$\therefore$  Phase difference,  $\phi = \frac{2\pi x}{\lambda}$  path difference.

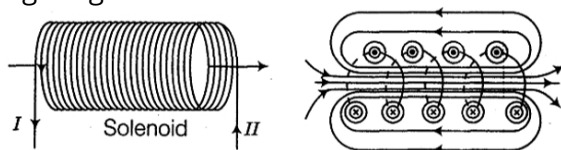
$$\text{So, } I_1 = I_0 \cos^2 \left( \frac{2\pi \times 0}{\lambda} \right) = I_0$$

$$\text{and } I_2 = I_0 \cos^2 \left( \frac{2\pi}{\lambda} \times \lambda/2 \right)$$

$$= I_0 \cos^2(\pi) = I_0$$

$$\therefore \frac{I_1}{I_2} = \frac{1}{1}$$

20. i. **Magnetic Field of a Solenoid** A long coil of wire consisting of closely packed loops is called solenoid, whose magnetic field resembles that of a bar magnet of south(S) and north(N) poles as shown in the figure given below.



Magnetic field of a solenoid

For points inside the core of toroid

Consider a circle of radius  $r$  in the region enclosed by turns of toroid. Now we apply Ampere's circuital law to this circular path, i. e.

$$\oint \vec{B} \cdot d\vec{l} = \mu I \dots (i)$$

$$\oint \vec{B} \cdot d\vec{l} = \oint B dl \cos 0 = B \cdot 2\pi r$$

Length of toroid =  $2\pi r$

Number of turns in toroid =  $n(2\pi r)$

current in one-turn =  $I$

$\therefore$  Current enclosed by circular path =  $(n2\pi r) \cdot I$

$\therefore$  Equation (1) gives

$$B \cdot 2\pi r = \mu_0 (n2\pi r I)$$

$$\Rightarrow B = \mu_0 n I$$

## ii. Toroid

A **toroid** is a hollow circular ring on which a large number of insulated turns of a metallic wire are closely wound. Solenoid and toroid are two equipments which are used to produce magnetic fields.

**Magnetic field inside the open space interior of the toroid** Let the loop 2 be shown in the figure, experience magnetic field  $B$ .

No current threads the loop 2 which lie in the open space inside the toroid.

$\therefore$  By Ampere's circuital law,

$$\oint_{\text{loop 2}} \vec{B} \cdot d\vec{l} = \mu_0 (I) = 0$$

$$\Rightarrow B = 0$$

**Magnetic field in the open space exterior of the toroid** Let us consider a coplanar loop 3 in the open space of exterior of toroid. Here, each turn of toroid threads the loop two times in opposite directions. Therefore, net current threading the loop

$$= NI - NI = 0$$

$\therefore$  By Ampere's circuital law,

$$\oint_{\text{loop 3}} B \cdot dl = \mu_0 (NI - NI) = 0$$

$$\Rightarrow B = 0$$

Thus, there is no magnetic field in the open space interior and exterior of the toroid.